3 Classification III: Invertebrates

Invertebrates are animals without internal backbones. They are usually less complex than vertebrates. There are over a million species of invertebrates. We can subdivide invertebrates into smaller groups depending on their characteristics. The groups we use are echinoderms, coelenterates, flatworms, segmented worms, molluscs and arthropods.

Echinoderms all live in the sea. They have a spiny skin. Starfish, sea urchins, sea cucumbers are all echinoderms.



Coelenterates nearly all live in the sea. They often have stinging cells. Jellyfish, hydra and sea anemones and coral are all coelenterates.

Flatworms are very simple creatures. They live in water or as parasites (live on other creatures). They all have flat bodies. Liver flukes, tapeworms and planaria are flatworms.

Segmented worms have long, thin and rounded bodies divided into segments (parts). Earthworms and leeches are segmented worms.

Molluscs mostly live in the sea or fresh water. They usually have shells. Many have a large 'foot'. Snails, octopuses, clams and oysters are molluscs.

Arthropods all have an exoskeleton (strong outside case). This is the largest group in the animal kingdom. Arthropod bodies are in sections and their legs are in pairs and jointed. Because it is such a large group, it is further divided into insects, arachnids, myriapods and crustaceans.

Insects have six legs. Their bodies are in three parts. Many insects have wings. The bee, housefly and ant are all insects.

Arachnids have eight legs and their bodies are divided into two parts. There are over 60 000 species of arachnids. As well as spiders this class includes scorpions, mites and ticks.

Myriapods The word myriapod means 'many legs'. The two main groups of myriapods are centipedes and millipedes. They both have a body with a head and a long trunk with many segments. Centipedes have one pair of legs per segment. Millipedes have two pairs of legs per segment.

Crustaceans have gills and antennae (feelers). Many are found in the sea. Lobsters, crabs, prawns and woodlice are all crustaceans.

1 Name the groups that make up invertebrates.

2 How many species of invertebrates are there in total?

3 What does a parasite do?

4 Give an example of a parasite.

5 Name two examples of an echinoderm.

6 Name two examples of a mollusc.

7 Name two examples of a flatworm.

8 What is an exoskeleton?

9 How many legs do arachnids have?

10 How can you tell centipedes and millipedes apart?

11 How would you classify yabbies? Give a reason for your answer.

12 Draw up a table to describe the characteristics of the four sorts of arthropods (insects, arachnids, myriapods and crustaceans).

13 Crustaceans are sometimes called 'insects of the sea'. Do you think that this is a appropriate name? Give a reason.

Research

14 In the class Myriapoda, one subclass is carnivorous, while the other is herbivorous. Find out which group (centipede or millipede) is the carnivore and which is the herbivore.

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